

Simulating a Weather Balloon to Minimize Cost

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I. Nomenclature

C_1	=	slope of temperature gradient
C_d	=	coefficient of drag
$\vec{F}_B$	=	buoyant force
$\vec{F}_D$	=	drag force
$\vec{F}_g$	=	force of gravity
$\vec{F}_{Net}$	=	net force
$\vec{F}_T$	=	tensile force
g	=	acceleration due to gravity
g_0	=	gravitational acceleration at sea level
G	=	shear modulus of natural latex
h	=	geopotential altitude
h_a	=	absolute altitude
h_G	=	geometric altitude
h_1	=	altitude at the base of the isothermal layer (11 km)
$m_{balloon}$	=	mass of balloon
m_{gas}	=	mass of the gas
$m_{hydrogen}$	=	mass of hydrogen
$m_{payload}$	=	mass of payload
M_{gas}	=	molar mass of the gas in the balloon
n	=	one mol of gas
$\hat{n}$	=	normal vector
P	=	pressure
P_0	=	pressure at sea level
P_{atm}	=	atmospheric pressure
$P_{elastic}$	=	pressure due to elastic strain
P_{in}	=	pressure inside the weather balloon
$\vec{r}$	=	position
r_0	=	initial balloon radius
R	=	ideal gas constant
R_E	=	radius of the Earth
t	=	thickness of the balloon
t_0	=	initial balloon thickness
T	=	temperature
T_0	=	temperature at sea level
T_1	=	temperature at 11 km altitude

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v	=	vertical velocity of the balloon relative to the air
V_{balloon}	=	volume of balloon
V_{gas}	=	volume of the gas in the balloon
z_0	=	initial height
λ	=	stretch ratio
ϕ	=	angle from the positive Z-axis
ρ_0	=	density of air at sea level
ρ_1	=	density of air at 11 km altitude
ρ_{air}	=	density of air
σ	=	stress
σ_C	=	critical stress
θ	=	angle in the XY-plane from the positive X-axis

II. Introduction

This report covers the analysis of how weather balloons operate and how changes within the atmosphere affect them. The importance of studying weather balloons is due to their impact when studying and measuring weather. Weather balloons are the main way that weather can be predicted and are an essential part of daily life. In recent years, the importance of weather balloons has increased due to climate change and the effects of global warming. These lead to global problems that need to be solved with data from weather balloons such as the ones highlighted in this report. The focus was on a hydrogen-filled balloon and a helium-filled balloon. Both balloons were compared in ascent and price to launch these balloons. The target height for these balloons was 35 kilometers. The balloons were simulated using Python and equations that were derived by hand. The goal was to see the amount of gas needed for each balloon to reach the targeted height and to analyze each of the balloon's paths to get there. Weather balloons were chosen to be studied because of the importance of hydrostatics in their applications. Hydrostatics was an important topic covered in the class and was thoroughly covered. With the exploration of hydrostatics, it was shown that local differences of pressure lead to a buoyant force, which is the main driver that pushes the weather balloons up. This idea of differences in pressure is extremely important for hydrostatics and many other topics covered in class.

III. Explanation of Real-World Problem

A. Introduction

Meteorologists use weather balloons every day to make accurate weather forecasts. A weather balloon consists of a large balloon, a parachute, and a payload that includes a small data collection device called a radiosonde. The balloon is filled with a gas that is less dense than air, most commonly helium or hydrogen, and is released from the ground [1]. The balloon uses the buoyancy force to ascend into the atmosphere, reaching as high as 130,000 feet before popping [2]. The physics of weather balloons directly corresponds to the study of hydrostatics in AE 311. This buoyancy force is the result of a pressure difference between the bottom and top of the balloon. However, it can be more intuitively understood as the force corresponding to the weight of the displaced fluid.

W_{disp} : Weight of displaced water P_{water} : Pressure of the water
 W_{block} : Weight of the block V_{disp} : Volume of the displaced water
 P_{atm} : Pressure of the atmosphere ρ_{water} : Density of the water

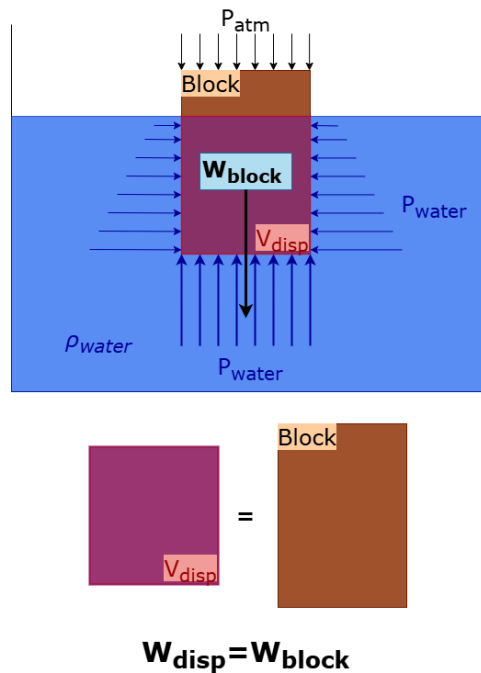


Fig. 1 Diagram demonstrating a block in hydrostatic equilibrium floating in water.

B. Generating Lift

For an object to be in hydrostatic equilibrium, it must reach a point in the fluid where the total weight of the object is equal to the weight of fluid displaced. This is shown by the block floating in the water in Fig. 1. To float, the weight of the water corresponding to the volume of the portion of the block submerged in water must equal the weight of the entire block. Only the volume of the block that is submerged is considered because that part physically displaces the water. The top of the block above the water is not responsible for moving any water out of the way. Given that the net buoyant force acts upward, the forces balance, leaving the block at equilibrium.

In the case of the weather balloon, the displaced fluid is now air. Given that gases occupy the volume they are contained in, the entire balloon is “submerged” in air. The gas inside the balloon is less dense than the surrounding air. Therefore, the weight of the air displaced by the inflated balloon is significantly greater than the weight of the gas inside the balloon, resulting in a net buoyant force greater in magnitude than the combined weight of the balloon and payload. This is what generates lift and allows the balloon to rise.

F_b : Force of buoyancy F_g : Weight of balloon and payload

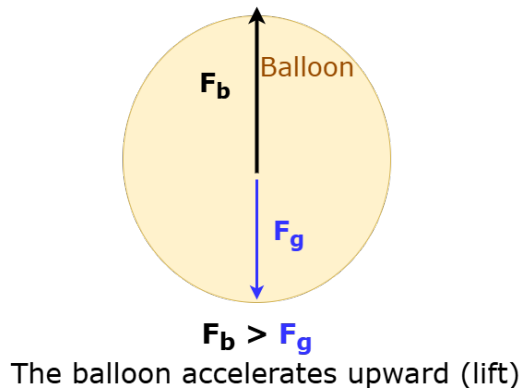


Fig. 2 Free body diagram of a weather balloon as it is released.

This lift exists for the balloon even when the balloon is at rest, but once the balloon begins to rise, it will experience a force of drag acting opposite to the balloon's motion. In reality, the drag on the balloon will vary as the balloon rises due to changes in the coefficient of drag, velocity, Reynolds number, and air density [3]. With all of these factors involved, it is incredibly difficult to accurately simulate the change in drag during the balloon's ascension. The drag coefficient on the weather balloon is assumed constant in the academic problem accordingly.

F_b : Force of buoyancy F_g : Weight of balloon and payload

F_d : Drag

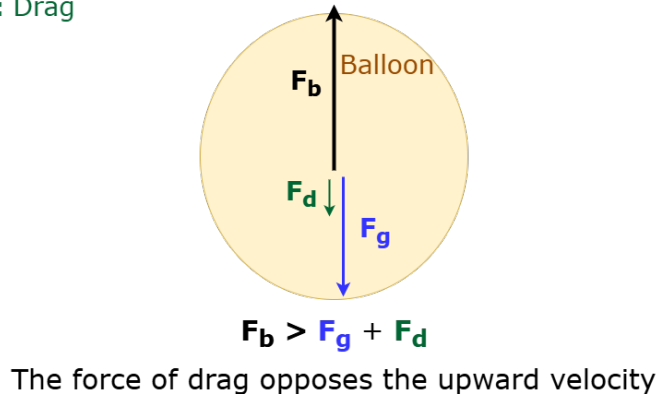


Fig. 3 Free body diagram of a weather balloon with a positive vertical velocity.

For the balloon gas, helium is the industry standard because it is not reactive and thus much safer to work with than hydrogen gas. However, helium is denser than hydrogen, costs much more per unit volume, and cannot be reliably manufactured [2]. In the academic problem, balloon trials are simulated with both hydrogen and helium, but hydrogen is the primary gas considered because of its lower cost and reliable access.

C. Flight of the Balloon

As the balloon rises, the radiosonde takes measurements of the air, including temperature, pressure, wind speed and direction, relative humidity, and altitude [4] [5]. It uses a battery powered radio transmitter to relay this information to a station on the ground [6]. The horizontal position of the balloon can also be measured to gather the wind direction and speed. The National Weather Service has 92 stations across the U.S. that release weather balloons twice a day to get comprehensive data spanning the entire country [5]. Data from the radiosondes is used by meteorologists to simulate weather prediction models, conduct research on weather and climate change, and create forecasts regarding severe weather, precipitation, pollution, and temperature [7].

Since air pressure decreases as altitude increases, the pressure around the balloon becomes less and less. As a result, the balloon begins to expand. It continually expands as it ascends until the balloon's material reaches its elastic limit [1]. Weather balloons are usually made out of latex, allowing them to expand far past their initial diameter. For example, a weather balloon with an initial diameter of 2.5 meters may expand to a diameter of up to 12.5 meters [1]. Any expansion

past the elastic limit causes the balloon to pop. Once the balloon pops, the parachute is deployed, and the radiosonde slowly descends back to Earth.

D. Complications with Data Acquisition and Proposed Solutions

Weather balloons can experience data contamination during ascent. This means that external factors cause data to be slightly inaccurate. These factors are overlooked in the academic problem, as they are difficult to model and account for due to their inconsistent nature. As the balloon ascends through the troposphere, which is the lowest level of the atmosphere, it expands nearly adiabatically. Adiabatic expansion means that the balloon expands without heat transfer across the balloon skin [8]. However, this changes when the balloon surpasses the tropopause – an intermediary layer between the troposphere and the stratosphere, the next highest layer of the atmosphere [9]. During the day, radiation from the sun heats the balloon, while at night heat transfers out of the balloon to the surrounding air. In either case, the radiosonde sits in the wake of this heat transfer, meaning that it will read artificially high or low temperatures during the day or night respectively [8].

Another unpredictable factor in data contamination is the buildup of humidity on the balloon skin. Moisture buildup is particularly prevalent in cloudy conditions since clouds are made up of water. Water droplets build up on the surface of the balloon as it rises. The balloon's surface also heats as it rises above the tropopause. This causes the water to evaporate. The wake of water vapor is picked up by the radiosonde's hygrometer (an instrument that measures humidity), therefore contaminating the data [8].

As the balloon rises, the payload can experience pendulum effects. This means that the payload may swing side-to-side or in circles while being lifted by the balloon [8]. The pendulum effects do not affect the upward motion of the balloon, allowing it to be safely neglected in the academic problem. In terms of data acquisition, pendulum motion is significant enough to impact measurement. Pendulum motion causes quick spikes in temperature, an effect that is exacerbated with greater altitude and decreased pressure [8]. This behavior is shown in Fig. 4:

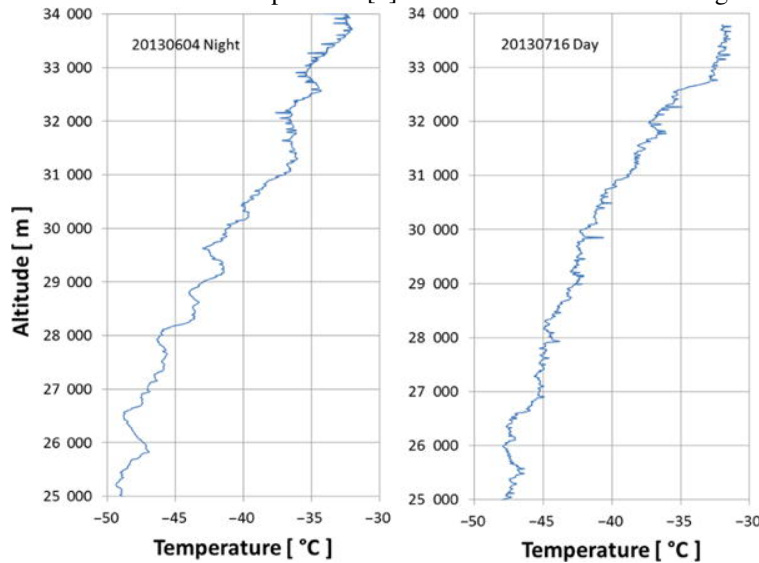


Fig. 4 Stratospheric temperature profiles measured from ascending balloons during nighttime (left panel) and daytime (right panel) with Meteolabor SRS-C34 [8].

Scientific balloons that measure radiation in the atmosphere are extra susceptible to pendulum motion because radiation sensors must remain as horizontal as possible. Hence, a swinging payload is highly detrimental to accurate radiation measurements [8].

Solutions to this contamination are largely directed toward stabilizing the descent of the weather balloon to get a greater volume of more accurate data. The NASA manuscript detailing the sources of data contamination presents two methods of controlled descent: the use of an automatic release valve on the balloon and using a 2 balloon system. The automatic release valve is tuned to open at a certain pressure roughly corresponding to the desired altitude for the balloon to reach. The trigger is set conservatively to help avoid premature bursts with the balloon. Once the set altitude is reached, the valve opens, releasing gas from inside the balloon out to the atmosphere. Since the balloon loses volume, it displaces less air and therefore has a buoyancy force lesser in magnitude than the balloon system's mass. From there,

the balloon begins its controlled descent to the earth [8].

The double balloon technique has a carrier balloon and a parachute balloon; it operates similarly to the valve technique by beginning the descent at a set altitude. The carrier balloon lifts the parachute balloon and payload up until the target altitude, where the carrier balloon is automatically detached from the rest of the system [8]. The parachute balloon does not provide a buoyancy force large enough to accelerate upward, meaning that the payload begins a controlled descent.

These systems are too complicated to enact at the scale of the National Weather Service, so they remain in use for higher precision scientific balloon launches requiring higher data accuracy. Given that the academic problem addressed in this report is tailored to the National Weather Service's needs, these techniques will not be employed nor addressed in the solution.

E. Connection to Course Content

As mentioned previously, the motion of a weather balloon is highly dependent on the properties of hydrostatics given that the only source of lift is the force of buoyancy on the balloon. Hydrostatics were thoroughly covered in class through lecture slides, example problems, and homework problems. More generally, weather balloons rely on pressure since hydrostatic balance is ultimately a balance of pressures. Pressure has been introduced in the course and will remain integral in all fluid mechanics topics covered in the course.

IV. Explanation of Academic-Style Problem

A. Academic Problem-Statement

You are looking to design a cost-effective weather balloon to be launched and collect data remotely. The balloon shall carry a payload of 3 kg and reach a minimum altitude of 35 km before bursting. You can assume that the balloon will be filled with hydrogen or helium gas, made of natural latex with an initial thickness of 0.2 mm, the balloon will maintain a spherical shape, and hydrogen and helium are ideal gases. The control variables will be the choice of gas and the quantity of gas used. How can you construct the balloon to accomplish the mission objective at the least cost?

B. Assumptions

The problem statement affords several assumptions, and a few other assumptions simplify the problem without meaningfully affecting the results.

- The balloon will remain spherical
- The balloon will be made of natural latex
- The balloon will have an initial thickness of 0.2 mm
- Hydrogen and helium are ideal gases
- The local pressure gradient along the balloon is a linear function of height.
- The Coefficient of Drag is constant for the balloon
- The payload has no effect on drag

C. Methods

To find the amount of gas necessary to lift the balloon, it is necessary to model the behavior of the balloon in the atmosphere. The overall approach used is a numerical integration of the acceleration of the balloon in the vertical basis. This acceleration directly results from the net force, according to Newton's Second Law. Using Euler's method, the acceleration can be numerically integrated over time to yield the vertical trajectory of the balloon.

At any moment in time, the balloon experiences the force of gravity, a buoyancy force due to the pressure gradient in the atmosphere, and a drag force acting opposite to the velocity of the balloon. At the moment the balloon is released, the velocity of the balloon is zero, therefore the drag force goes to zero. Along the balloon's ascent, even if the drag force is enough to slow down the balloon, any buoyant force which exceeds the force of gravity will continue to propel the balloon upward. For this reason, finding the buoyant force is essential to calculate the maximum altitude of the balloon. This force can be found by integrating the pressure of the balloon over the surface of the balloon. In this pursuit, it is useful to assume a linear pressure gradient. The atmospheric model will show this to be false over kilometers of altitude, but, in the local perspective, the change in pressure can be closely approximated as a function of height.

It will be shown that the buoyant force is a function of the volume of the balloon and the density of the air. Both of these vary as the balloon ascends through the atmosphere due to drops in atmospheric pressure and temperature changes. Therefore, a model of the atmosphere will be essential in the development of a meaningful model of the balloon's behavior [10]. The size of the balloon is also dependent on the material properties of its construction, and furthermore the material comprising the balloon will dictate the critical stress necessary to cause failure [11], so it is also critical to understand the material characteristics of the balloon. Finally, to determine the cost of the gas comprising the balloon, bulk prices were found to give a standard baseline for the cost of each gas.

The python program included in the appendix is used to generate plots modeling the standard atmosphere of the balloon and the vertical trajectory of the balloon for a successful launch, and to calculate the minimum quantity of gas used for a user input type of gas. It works by first using if/then statements to create piecewise functions of pressure, temperature, and density for the region of atmosphere the balloon may occupy, then defines Euler's method with the derived acceleration to define the path over a maximum range of time. At each iteration of this method, the stress calculated in the material is compared to the critical stress for the material to determine if the balloon has burst. A simulation block runs this method for a base input value and iteratively increases the mass of gas used in the balloon until the success criterion is met. This iterative process is stored and output to a csv file, the final row of which contains the solution to the least-cost quantity for that type of gas.

1. Finding Buoyant Force

$$\vec{F}_B = \int_S P(z) \hat{n} dA \quad (1)$$

Defining the $P(z)$:

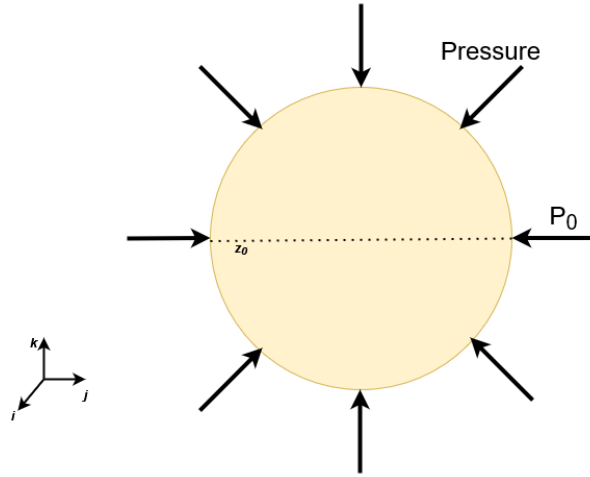


Fig. 5 Diagram showing the pressure distribution over the balloon.

Pressure is a function of altitude; in this case, the local pressure difference from a reference pressure will be assumed to be linearly related to the height above it by a factor $\rho_{\text{air}}g$.

$$\begin{aligned} P(z) &= P_0 + \rho_{\text{air}}gz \\ z &= z_0 + r \cos \phi \\ P(z) &= P_0 + \rho_{\text{air}}g(z_0 + r \cos \phi) \end{aligned}$$

Finding the Normal Vector, $\hat{n}$:

$$\begin{aligned} \vec{r} &= \langle r, \theta, \phi \rangle \\ r &= r_0 \end{aligned}$$

$$\vec{r}(\theta, \phi) = r_0 \sin \phi \cos \theta \hat{i} + r_0 \sin \phi \sin \theta \hat{j} + r_0 \cos \phi \hat{k} \quad (2)$$

$$0 \leq \theta \leq 2\pi \quad 0 \leq \phi \leq \pi$$

$$\vec{r}_\theta(\theta, \phi) = -r_0 \sin \theta \sin \phi \hat{i} + r_0 \sin \phi \cos \theta \hat{j}$$

$$\vec{r}_\phi(\theta, \phi) = r_0 \cos \theta \cos \phi \hat{i} + r_0 \cos \phi \sin \theta \hat{j} - r_0 \sin \phi \hat{k}$$

$$\hat{n} = \frac{\vec{r}_\theta \times \vec{r}_\phi}{|\vec{r}_\theta \times \vec{r}_\phi|}$$

$$\vec{r}_\theta \times \vec{r}_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -r_0 \sin \phi \sin \theta & r_0 \sin \phi \cos \theta & 0 \\ r_0 \cos \phi \cos \theta & r_0 \cos \phi \sin \theta & -r_0 \sin \phi \end{vmatrix} = -r_0^2 \sin^2 \phi \cos \theta \hat{i} - r_0^2 \sin^2 \phi \sin \theta \hat{j} - r_0^2 \sin \phi \cos \phi \hat{k}$$

$$\begin{aligned} |\vec{r}_\theta \times \vec{r}_\phi| &= \sqrt{r_0^4 \sin^4 \phi \cos^2 \theta + r_0^4 \sin^4 \phi \sin^2 \theta + r_0^4 \sin^2 \phi \cos^2 \phi} \\ &= \sqrt{r_0^4 \sin^4 \phi (\cos^2 \theta + \sin^2 \theta) + r_0^4 \sin^2 \phi \cos^2 \phi} \\ &= \sqrt{r_0^4 \sin^2 \phi (\sin^2 \phi + \cos^2 \phi)} = r_0^2 \sin \phi \end{aligned}$$

$$\hat{n} = \frac{-r_0^2 \sin^2 \phi \cos \theta \hat{i} - r_0^2 \sin^2 \phi \sin \theta \hat{j} - r_0^2 \sin \phi \cos \phi \hat{k}}{r_0^2 \sin \phi} = -\sin \phi \cos \theta \hat{i} - \sin \phi \sin \theta \hat{j} - \cos \phi \hat{k}$$

This vector points towards the center of the balloon, while the desired vector, $\hat{n}$, points outward from the volume of the balloon.

$$\hat{n} = \sin \phi \cos \theta \hat{i} + \sin \phi \sin \theta \hat{j} + \cos \phi \hat{k} \quad (3)$$

The integral will be simpler to evaluate over the spherical balloon using a spherical coordinate system. This is why the normal vector and the pressure have been defined in terms of their spherical components. By definition,

$$dA = r^2 \sin \theta d\theta d\phi$$

Putting it all together,

$$\vec{F}_B = \int_0^\pi \int_0^{2\pi} [P_0 + \rho_{\text{air}} g(z_0 + r \cos \phi)] (\sin \phi \cos \theta \hat{i} + \sin \phi \sin \theta \hat{j} + \cos \phi \hat{k}) r^2 \sin \phi d\theta d\phi \quad (4)$$

$$\vec{F}_B = \int_0^\pi \int_0^{2\pi} (P_0 + \rho_{\text{air}} g z_0 + \rho_{\text{air}} g r \cos \phi) (\sin \phi \cos \theta \hat{i} + \sin \phi \sin \theta \hat{j} + \cos \phi \hat{k}) r^2 \sin \phi d\theta d\phi$$

Distributing the terms on the left side of the integral provides an opportunity to separate the integral into three parts.

Integrating the P_0 term,

$$\begin{aligned}
& \int_0^\pi \int_0^{2\pi} P_0 (\sin \phi \cos \theta \hat{i} + \sin \phi \sin \theta \hat{j} + \cos \phi \hat{k}) r^2 \sin \phi d\theta d\phi \\
&= P_0 r^2 \int_0^\pi \int_0^{2\pi} \sin^2 \phi \cos \theta \hat{i} + \sin^2 \phi \sin \theta \hat{j} + \sin \phi \cos \phi \hat{k} d\theta d\phi \\
&= P_0 r^2 \int_0^\pi \left(\sin \phi \sin \theta \hat{i} - \sin^2 \phi \cos \theta \hat{j} + \theta \sin \phi \cos \phi \hat{k} \right) \Big|_0^{2\pi} d\phi \\
&= P_0 r^2 \int_0^\pi (\sin^2 \phi (0 - 0) \hat{i} - \sin^2 \phi (1 - 1) \hat{j} + 2\pi \sin \phi \cos \phi \hat{k}) d\phi \\
&= P_0 r^2 \int_0^\pi 2\pi \sin \phi \cos \phi \hat{k} d\phi = P_0 r^2 \int_0^\pi \pi \sin 2\phi \hat{k} d\phi \\
&= P_0 r^2 \left(-\frac{\pi}{2} \cos 2\phi \right) \Big|_0^\pi \hat{k} = P_0 r^2 \left(-\frac{\pi}{2} (1 - 1) \right) \hat{k} = 0
\end{aligned}$$

This means that absolute pressure does not affect the buoyancy force, only changes in pressure.

Integrating the $\rho_{\text{air}} g z_0$ term,

$$\rho_{\text{air}} g z_0 \int_0^\pi \int_0^{2\pi} (\sin \phi \cos \theta \hat{i} + \sin \phi \sin \theta \hat{j} + \cos \phi \hat{k}) r^2 \sin \phi d\theta d\phi = \rho_{\text{air}} g z_0 (0) = 0$$

Integrating the $\rho_{\text{air}} g r \cos(\phi)$ term,

$$\begin{aligned}
& \int_0^\pi \int_0^{2\pi} (\rho_{\text{air}} g r \cos \phi) (\sin \phi \cos \theta \hat{i} + \sin \phi \sin \theta \hat{j} + \cos \phi \hat{k}) r^2 \sin \phi d\theta d\phi \\
&= \rho_{\text{air}} g r^3 \int_0^\pi \int_0^{2\pi} (\sin^2 \phi \cos \phi \cos \theta \hat{i} + \sin^2 \phi \cos \phi \sin \theta \hat{j} + \cos^2 \phi \sin \phi \hat{k}) d\theta d\phi \\
&= \rho_{\text{air}} g r^3 \int_0^\pi (\sin^2 \phi \cos \phi \sin \theta \hat{i} - \sin^2 \phi \cos \phi \cos \theta \hat{j} + \theta \cos^2 \phi \sin \phi \hat{k}) \Big|_0^{2\pi} d\phi \\
&= \rho_{\text{air}} g r^3 \int_0^\pi \sin^2 \phi \cos \phi (0 - 0) \hat{i} - \sin^2 \phi \cos \phi (1 - 1) \hat{j} + (2\pi - 0) \cos^2 \phi \sin \phi \hat{k} d\phi \\
&= -\rho_{\text{air}} g r^3 \int_0^\pi -2\pi \cos^2 \phi \sin \phi \hat{k} d\phi \\
&= -2\pi \rho_{\text{air}} g r^3 \hat{k} \int_0^\pi -\cos^2 \phi \sin \phi d\phi
\end{aligned}$$

Using U-Substitution, with $u = \cos \phi$ and $du = -\sin \phi d\phi$

$$\int_0^\pi -\cos^2 \phi \sin \phi d\phi = \int_1^{-1} u^2 du = \frac{1}{3} u^3 \Big|_1^{-1} = \frac{1}{3} (-1)^3 - \frac{1}{3} (1)^3 = -\frac{2}{3}$$

Final Buoyancy Force Equation

$$\begin{aligned}\vec{F}_B &= -2\pi\rho_{\text{air}}gr^3\hat{k}\left(-\frac{2}{3}\right) = \frac{4}{3}\pi r^3\rho g\hat{k} \\ \vec{F}_B &= \frac{4}{3}\pi r^3\rho g\hat{k}\end{aligned}$$

$$\vec{F}_B = V_{\text{balloon}}\rho_{\text{air}}g\hat{k} \quad (5)$$

2. Finding Net Force

$$\vec{F}_{\text{Net}} = \vec{F}_g + \vec{F}_B + \vec{F}_D \quad (6)$$

$$\vec{F}_g = -(m_{\text{balloon}} + m_{\text{hydrogen}} + m_{\text{payload}})g\hat{k}$$

Drag on a perfect sphere:

$$\vec{F}_D = -\frac{1}{2}C_d\rho_{\text{air}}v^2A$$

$$\vec{F}_B = \rho_{\text{air}}V_{\text{balloon}}g\hat{k}$$

$$\vec{F}_{\text{Net}} = \left[(\rho_{\text{air}}V_{\text{balloon}} - m_{\text{balloon}} - m_{\text{hydrogen}} - m_{\text{payload}})g - \frac{1}{2}C_d\rho_{\text{air}}V^2A \right] \hat{k} \quad (7)$$

3. Standard Atmosphere Model

To model the behavior of a weather balloon over time, one must know how the properties of the atmosphere vary with respect to altitude. In this pursuit, a standard atmospheric model has been developed. The temperature varies linearly with altitude in gradient regions and remains constant in isothermal regions of the atmosphere. Any other properties of interest, including pressure and density, can be modeled with the equations below from known reference values and relative temperature changes. Furthermore, the diameter of a balloon can be modeled as a function of altitude because the change in size of a balloon released from sea level depends only on the ambient temperature and atmospheric pressure at the current altitude.

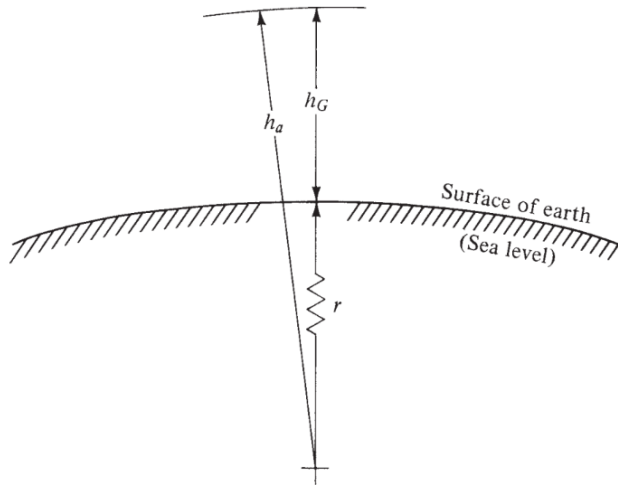


Fig. 6 Standard atmosphere model of altitude[10].

$$h = \frac{R_E}{R_E + h_G} h_G$$

Solving for h_G ,

$$h_G = \frac{h R_E}{R_E - h}$$

Acceleration due to gravity varies with altitude

$$g = g_0 \left(\frac{R_E}{h_a} \right)^2 = g_0 \left(\frac{R_E}{R_E + h_G} \right)$$

$$g_0 = 9.80665 \text{ m/s}^2$$

Pressure

$$\frac{dP}{P} = -\frac{\rho_{\text{air}} g_0 dh}{\rho_{\text{air}} R T} = -\frac{g_0}{R T} dh$$

The temperature varies linearly as the altitude increases, passing through gradient regions and isothermal regions. The target altitude for this problem is 35 km, so the model must include nearby regions.

- Negative gradient region from 0 km $\rightarrow$ 11 km
- Isothermal region from 11 km $\rightarrow$ 25 km
- Positive gradient region from 25 km $\rightarrow$ 47 km

This model uses known reference values at the lower bounds of these layers to calculate the relative pressure and density changes relative to changes in temperature.

Gradient Layers

$$C_1 = \frac{dT}{dh} \Rightarrow \frac{P}{P_1} = \left(\frac{T}{T_1} \right)^{-\frac{g_0}{C_1 R}}$$

$$\frac{P}{P_1} = \frac{\rho_{\text{air}} T}{\rho_1 T_1} \Rightarrow \frac{\rho_{\text{air}}}{\rho_1} = \left(\frac{T}{T_1} \right)^{-\frac{g_0}{C_1 R} - 1}$$

$$\text{Where } T = T_1 + C_1(h - h_1)$$

at $h = 0$

$$\begin{cases} P_0 = 101,325 \text{ Pa} \\ \rho_0 = 1.2250 \text{ kg/m}^3 \\ T_0 = 288.16 \text{ K} \end{cases}$$

Isothermal layer

$$\frac{P}{P_1} = e^{-\left(\frac{g_0}{RT}\right)(h-h_1)} = \frac{\rho_{\text{air}}}{\rho_1}$$

h_1 is the altitude at the base of the isothermal layer, h is some altitude in the isothermal layer

Equation for stress for a thin-walled sphere[11]

$$\sigma = \frac{(P_{\text{in}} - P_{\text{atm}})r}{2t} = \frac{\Delta P_{\text{elastic}} r}{2t} \quad (8)$$

$$P_{\text{in}} = P_{\text{atm}} + \Delta P_{\text{elastic}} \quad (9)$$

Change in pressure between inside and outside of the balloon due to the elastic material of the balloon,

$$\Delta P_{\text{elastic}} = P_{\text{in}} - P = \frac{2F_T}{r}$$

$$F_T = \sigma t \rightarrow \Delta P_{\text{elastic}} = \frac{2\sigma t}{r}$$

Thickness Equation

$$t = t_0 \left(\frac{r_0}{r} \right)^2$$

Stretch ratio

$$\lambda = \frac{r}{r_0} \Rightarrow t = \frac{t_0}{\lambda^2}$$

$$\sigma = G \left(\lambda - \frac{1}{\lambda^2} \right)$$

$$\Delta P_{\text{elastic}} = \frac{2t}{r} \left[G \left(\lambda - \frac{1}{\lambda^2} \right) \right] = \frac{2t_0 G}{r \lambda^2} \left(\lambda - \frac{1}{\lambda^2} \right) = \frac{2t_0 G}{r} \left(\frac{r_0}{r} - \frac{r_0^4}{r^4} \right) = 2t_0 G \left(\frac{r_0}{r^2} - \frac{r_0^4}{r^5} \right) \quad (10)$$

Finding the material stress

$$\sigma = \frac{\Delta P r}{2t} = \frac{r}{2t} \left[2t_0 G \left(\frac{r_0}{r^2} - \frac{r_0^4}{r^5} \right) \right] = \frac{G t_0 r^3}{t_0 r_0^2} \left(\frac{r_0}{r^2} - \frac{r_0^4}{r^5} \right) = G \left(\frac{r}{r_0} - \frac{r_0^2}{r^2} \right)$$

Bursting Stress

$$\sigma_c = G \left(\frac{r}{r_0} - \frac{r_0^2}{r^2} \right) \quad (11)$$

Temperature Dependence of Criterion

$$\sigma_c(T) = \sigma_{c,0} [1 - 0.01(T - T_0)]$$

Finding a function for the radius of the balloon

$$\Delta P_{\text{elastic}} = P_{\text{in}} - P_{\text{atm}}$$

$$\Delta P_{\text{elastic}} = \frac{2Gt}{r} \left(\frac{r}{r_0} - \frac{r_0^2}{r^2} \right) = \frac{2Gt_0 r_0^2}{r^3} \left(\frac{r}{r_0} - \frac{r_0^2}{r^2} \right) = 2Gt_0 \left(\frac{r_0}{r^2} - \frac{r_0^4}{r^5} \right)$$

$$P_{\text{in}} V_{\text{gas}} = nRT$$

$$V_{\text{gas}} = \frac{4}{3} \pi r^3 \quad (12)$$

$$P_{\text{in}} = \frac{nRT}{V_{\text{gas}}} = \frac{3nRT}{4\pi r^3} \quad (13)$$

- R = Ideal Gas Constant
- n = mol of gas = $\frac{m_{\text{gas}}}{M_{\text{gas}}}$
- M_{gas} = molar mass of the gas

$$\begin{aligned}
2Gt_0 \left(\frac{r_0}{r^2} - \frac{r_0^4}{r^5} \right) &= \frac{3nRT}{4\pi r^3} - P_{\text{atm}} \\
\frac{2Gt_0 r_0}{r^2} - \frac{2Gt_0 r_0^4}{r^5} &= \frac{3nRT}{4\pi r^3} - P_{\text{atm}} \\
8\pi Gt_0 r_0 r^3 - 8\pi Gt_0 r_0^4 &= 3nRT r^2 - 4\pi r^5 P_{\text{atm}} \\
4\pi P_{\text{atm}} r^5 + 8\pi Gt_0 r_0 r^3 - 3nRT r^2 - 8\pi Gt_0 r_0^4 &= 0
\end{aligned} \tag{14}$$

Finding the initial radius

$$\begin{aligned}
4\pi P_{\text{atm}} (r_0)^5 + 8\pi Gt_0 r_0 (r_0)^3 - 3nRT (r_0)^2 - 8\pi Gt_0 r_0^4 &= 0 \\
4\pi P_{\text{atm}} r_0^5 - 3nRT r_0^2 &= 0 \\
4\pi P_{\text{atm}} r_0^3 - 3nRT &= 0 \\
r_0 &= \sqrt[3]{\frac{3nRT}{4\pi P_{\text{atm}}}}
\end{aligned} \tag{15}$$

D. Results and Discussion

1. The Atmospheric Model

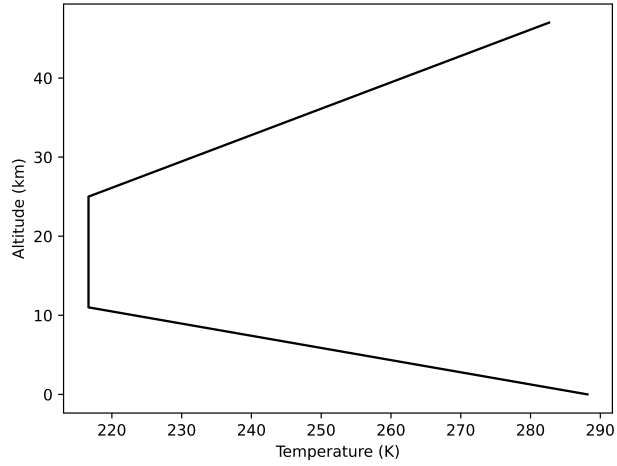


Fig. 7 The ambient temperature in Kelvin versus the altitude in kilometers for the atmospheric model.

As the altitude increases, the ambient temperature drops linearly with respect to an increase in altitude up to 11 kilometers. From 11 kilometers to 25 kilometers of altitude, the temperature remains constant, and from 25 kilometers to 47 kilometers of altitude, the temperature linearly increases with increasing altitude.

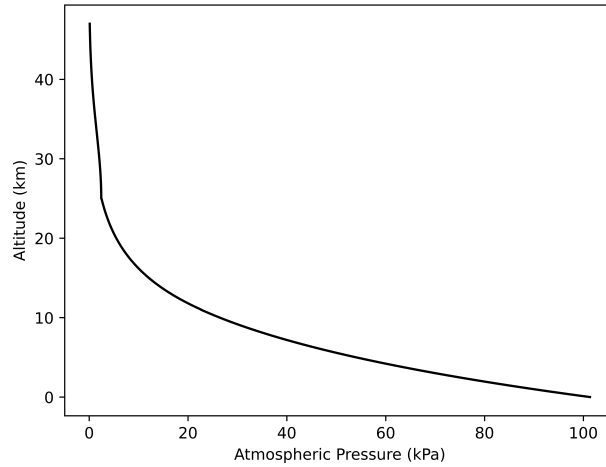


Fig. 8 The atmospheric pressure in KPa versus the altitude in kilometers for the atmospheric model.

The pressure exponentially decreases with increasing altitude across the entire range of altitudes.

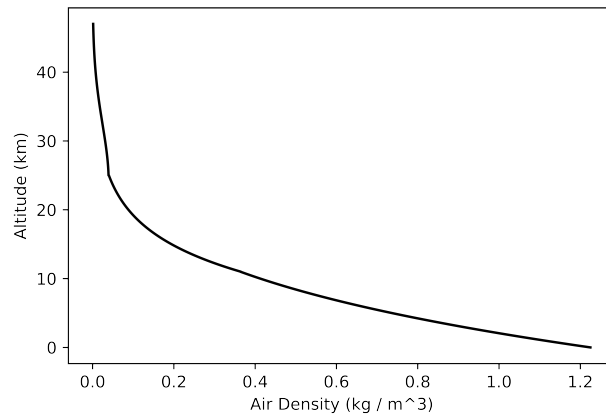


Fig. 9 The air density in kg / m³ versus the altitude in kilometers for the atmospheric model.

Similar to pressure, the air density decreases exponentially with increasing altitude.

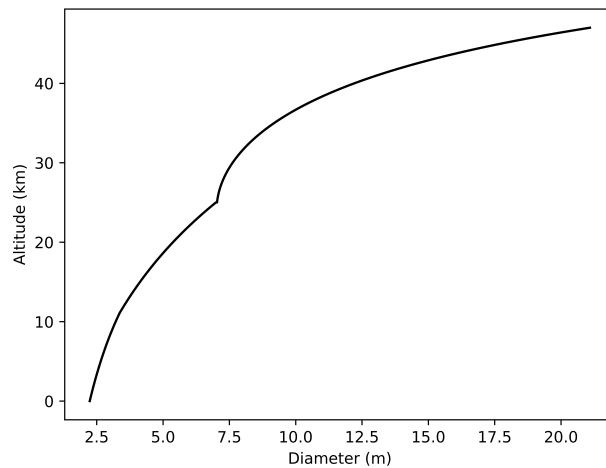


Fig. 10 The diameter of a balloon in meters versus the altitude in kilometers for the atmospheric model.

This plot shows how a hypothetical balloon with an initial diameter would behave at different altitudes. The most appropriate intuition for this graph is to imagine a balloon that is allowed to expand to a stable equilibrium size at each altitude on the plot. The diameter of this balloon increases at a decreasing rate with increasing altitude.

2. Simulation Results

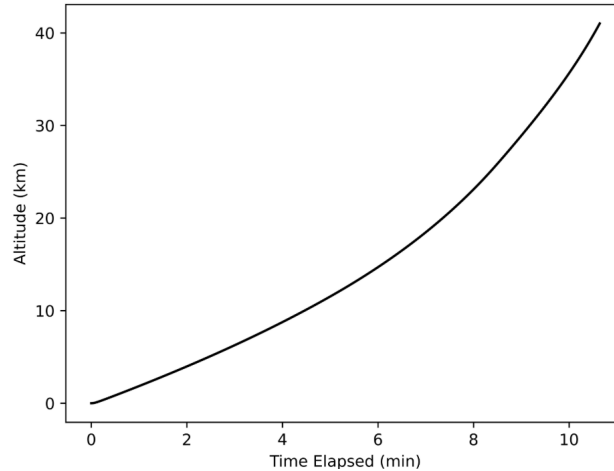


Fig. 11 The vertical trajectory of the simulated balloon in kilometers versus against the time elapsed in minutes.

The trajectory of the first successful balloon to reach the target altitude in the iterative process is shown. The altitude increases at an increasing rate as time elapses.

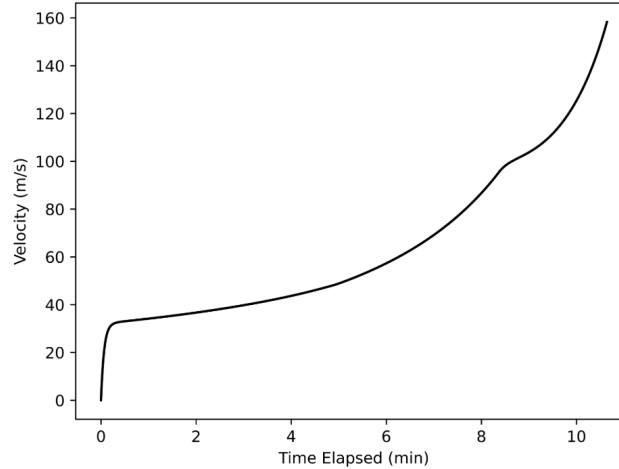


Fig. 12 The velocity of the simulated balloon in meters per second versus the time elapsed in minutes.

The velocity of the first successful balloon to reach the target altitude in the iterative process is shown. The velocity increases rapidly at first due to a large net buoyancy force, then flattens out due to an increase in drag force acting against the direction of movement, and finally increases at an increasing rate for most of the remaining altitude.

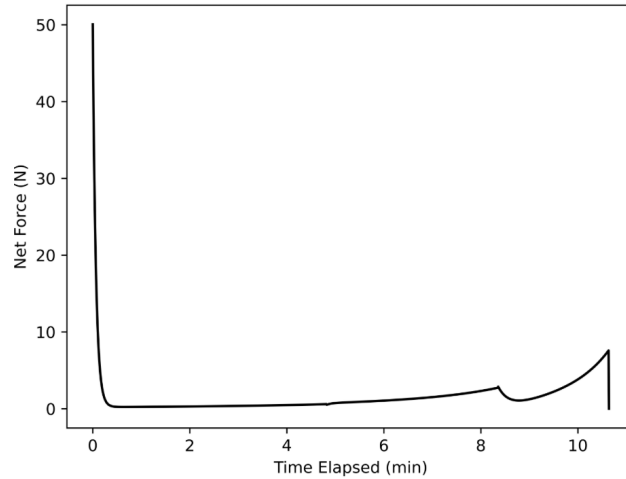


Fig. 13 The net force on the balloon in Newtons versus the time elapsed in minutes.

A positive net force is defined to act in the $+\hat{j}$ direction. The net force initially reflects a large net buoyancy force, until the drag force increases to roughly equal this buoyancy force. From this point, the net force slowly increases for the duration of the flight.

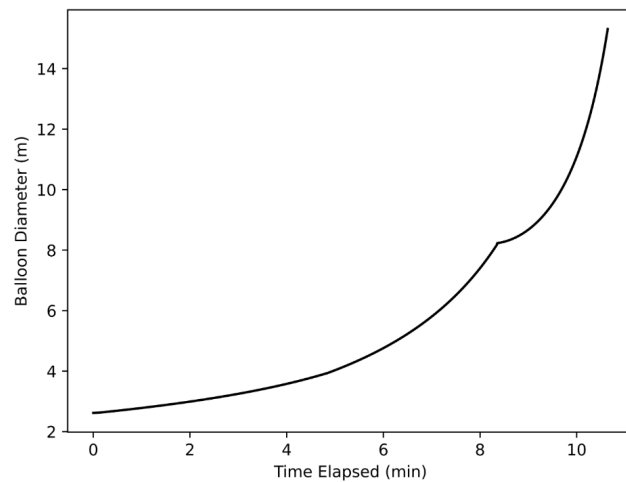


Fig. 14 The diameter of the simulated balloon in meters versus the time elapsed in minutes.

The diameter of the balloon roughly increases at an increasing rate over the course of the flight duration.

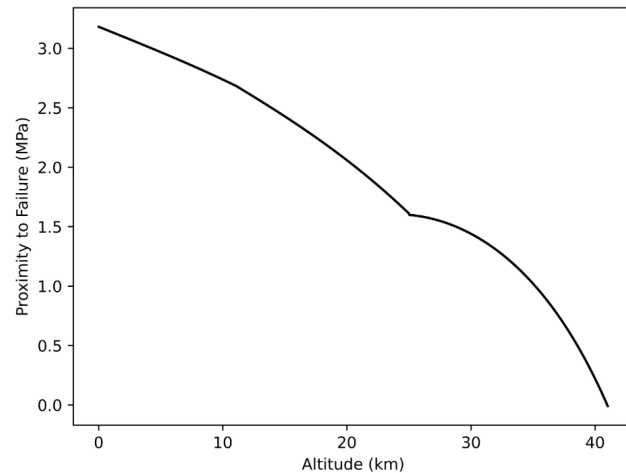


Fig. 15 The proximity to material failure of the simulated balloon in MPa versus the altitude in kilometers.

The balloon's proximity to material failure, or bursting, decreases exponentially with increasing altitude.

3. Data Tables

m_{H_2} (kg)	Cost(\$USD)	t_0 (mm)	D (m)	a (km)	success?
0.20	0.97	0.2	1.6	0.0	fail
0.21	1.01	0.2	1.7	0.0	fail
0.22	1.06	0.2	1.7	0.0	fail
0.23	1.11	0.2	1.7	0.0	fail
0.24	1.16	0.2	1.7	0.0	fail
0.25	1.21	0.2	1.8	0.0	fail
0.26	1.25	0.2	1.8	0.0	fail
0.27	1.30	0.2	1.8	0.0	fail
0.28	1.35	0.2	1.8	0.0	fail
0.29	1.40	0.2	1.9	0.0	fail
0.30	1.45	0.2	1.9	0.0	fail
0.31	1.50	0.2	1.9	0.0	fail
0.32	1.54	0.2	1.9	0.0	fail
0.33	1.59	0.2	1.9	0.0	fail
0.34	1.64	0.2	2.0	0.0	fail
0.35	1.69	0.2	2.0	0.0	fail
0.36	1.74	0.2	2.0	0.0	fail
0.37	1.79	0.2	2.0	0.0	fail
0.38	1.83	0.2	2.0	0.0	fail
0.39	1.88	0.2	2.1	0.0	fail
0.40	1.93	0.2	2.1	0.0	fail
0.41	1.98	0.2	2.1	0.0	fail
0.42	2.03	0.2	2.9	8.93	fail
0.43	2.07	0.2	14.5	43.12	success

Table 1 Table showing the relation between the mass of hydrogen gas put in the balloon in kilograms, m_{H_2} , the cost of the gas in the balloon in \$USD, the initial thickness of the balloon in millimeters, t_0 , the final diameter of the balloon in meters, D , the highest altitude reached by the balloon in kilometers, a , and if the balloon succeeded or failed the mission objective for each trial as hydrogen gas is added to the balloon between trials.

The table shows the iteration process executed by the code to add gas to the balloon until the success criterion is met. The final row contains the necessary quantity of hydrogen gas, the price of that quantity, the thickness of the balloon, the final diameter of the balloon, and the maximum altitude reachable by the balloon.

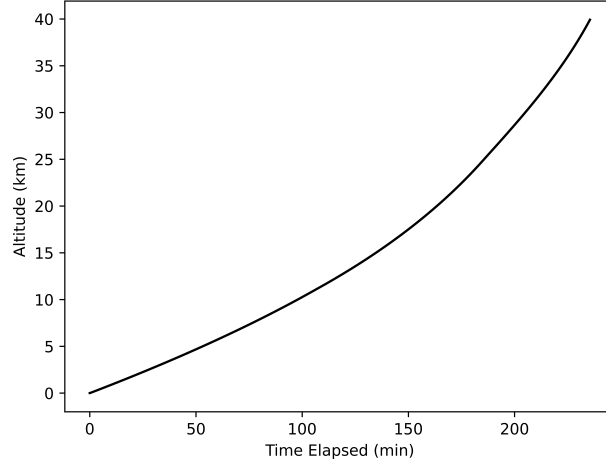


Fig. 16 The trajectory of a simulated balloon filled with helium in kilometers versus the time elapsed in minutes.

The Helium-filled balloon follows a similar trend to the hydrogen filled balloon. The altitude of the balloon increases at an increasing rate as the time elapsed increases. In this case, however, the time elapsed to reach the target altitude is far greater.

m_{He} (kg)	Cost(\$USD)	t_0 (mm)	D (m)	a (km)	success?
0.80	76.1	0.2	2.1	0.0	fail
0.81	77.05	0.2	2.1	0.0	fail
0.82	78.0	0.2	2.1	0.0	fail
0.83	78.95	0.2	2.1	0.0	fail
0.84	79.9	0.2	2.1	0.0	fail
0.85	80.85	0.2	2.1	0.0	fail
0.86	81.8	0.2	2.1	0.0	fail
0.87	82.75	0.2	2.1	0.0	fail
0.88	83.71	0.2	2.1	0.0	fail
0.89	84.66	0.2	2.2	0.0	fail
0.90	85.61	0.2	2.2	0.0	fail
0.91	86.56	0.2	1.9	0.0	fail
0.92	87.51	0.2	1.9	0.0	fail
0.93	88.46	0.2	2.7	6.29	fail
0.94	89.41	0.2	4.6	17.39	fail
0.95	90.36	0.2	11.4	39.14	success

Table 2 Table showing the relation between the mass of helium gas put in the balloon in kilograms, m_{H_2} , the cost of the gas in the balloon in \$USD, the initial thickness of the balloon in millimeters, t_0 , the final diameter of the balloon in meters, D , the highest altitude reached by the balloon in kilometers, a , and if the balloon succeeded or failed the mission objective for each trial as helium gas is added to the balloon between trials.

The table shows the iteration process executed by the code to add gas to the balloon until the success criterion is met, this time for Helium. The final row contains the necessary quantity of helium gas, the price of that quantity, the thickness of the balloon, the final diameter of the balloon, and the maximum altitude reachable by the balloon.

4. Interpreting the final results

In this investigation, the primary objective was to find the cheapest type and quantity of gas needed to inflate a 0.2 mm thickness weather balloon to reach a target altitude of 35 kilometers. The simulations show that the quantity of hydrogen needed to reach the target altitude was 0.43 kilograms. At a price of 4.85 \$USD, the cost to use hydrogen would be \$2.07 [12]. The quantity of helium necessary was 0.95 kilograms. The price of helium is 95.12 \$USD, making the cost of helium \$90.36 [13]. Helium's greater molar mass and higher price point both drive up the cost compared to hydrogen. A hydrogen-filled balloon is less costly than a helium filled balloon for this situation.

V. Connection between the Real-World problem, the Academic-Style problem, and the course content

The transition between the real world problem and academic style problem was vital when analyzing how weather balloons behave in the real world and how it was simulated within this report. When considering the balloon in real life it is important to take into consideration some aspects that would have an effect on how the balloon ascends.

The most important similarity between the two problems is the presence of a buoyancy force in creating lift. It is the only source of lift for a weather balloon and, therefore, necessary to accurately simulate a weather balloon's motion. Therein also lies the primary connection to the course content. The force of buoyancy is categorized under the study of hydrostatics, which was covered in the class. The two problems are also similar in how the balloon expands until eventually popping. Since the Standard Atmosphere Model was used in the simulation, pressure decreased as altitude increased. The expansion of the balloon up until it reached its critical stress was considered. This meant that the balloon's ascent ended when it popped, just as it would in the real world. Cost considerations are also highlighted in both problems. The minimization of cost was a key driver in the academic problem statement, and companies such as the National Weather Service must always factor in cost into their logistics to successfully run the organization.

One concept that was different between the academic and real life problem was that pressure was assumed to change linearly throughout the volume of the balloon. This assumption was also made in lecture to make hydrostatics problems relatively straightforward to solve, but this is not the case in real life. Another assumption was that the balloon would remain a perfect sphere throughout its ascent. Tangentially, the drag coefficient on the balloon was assumed constant. In reality, the balloon is not rigid which allows the balloon to change shape, altering the drag coefficient in the process. While the Standard Atmosphere Model is a reputable model of the Earth's atmosphere, it does not factor wind into the model. Any model incorporates assumptions into its structure, representing another deviation from the real world.

VI. Team Management

In group discussions on how the work would be distributed, it was decided that the work would be separated according to the strengths of each member. McDonald was the one who came up with the idea for the problem, so he was assigned to lead the Real-World problem section. Lutter had experience working with weather balloons through a program at Rice University, so he was given the position of Technical lead. Rees was then assigned the role of team leader and communication because of his interest in both areas of the problem and organizational skills.

A. Tasks Assigned

The tasks that were assigned to each position were based on the deliverables of the overall problem. The Real-World problem lead was given the task of correctly defining the problem and its relation to the class. We wanted to focus on a problem that dealt with the applications of hydrostatics. The Real-World problem Lead ended up deciding on choosing a weather balloon due to the importance of hydrostatics and buoyancy forces with its applications. He was then tasked with researching and documenting what was currently being used and how it could help solve our overall problem. After his research, he created a general solution of how the problem was going to be solved. He wanted to base our balloon on the common weather balloons used by the National Weather Service. He continued his research by looking at alternate solutions that were too complicated for the academic problem. These other solutions helped create depth and a better understanding of the overall problem and its components. The Technical lead was tasked with the responsibility of taking the problem from the Real-World problem lead and creating an exam-style solution for it. When figuring out the solution, one has to make different assumptions so that the problem can be solved using the methods that were taught in class. He also needed to ensure that the academic problem was difficult enough. Another of his tasks was to ensure that the solution was correct. He did so by simulating his findings within Python. He graphed the results

in different plots that confirmed the ideas of the Real-World problem lead. He was also tasked with documenting all his findings in a clear and readable manner. The Team Manager and Communicator was tasked with being a lead and the connector between the Real-World problem lead and the Technical lead. The Team Manager was also tasked with the organization of the group and assigning each member their specific tasks and division of labor. He also created meeting times and an overall schedule that the group should follow. He was responsible for setting specific deadlines and overall team management. When working on the final draft of the document the Team Manager was tasked with outlining the final draft and making sure that the team follows AIAA formatting.

B. Project Schedule

AE 311 Weather Balloon Schedule								
	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Weekly Goals
Week of 2/10/2025	Team Meeting 1:00 Pm		Office hours 3:00 Pm - 5:00 Pm Team Meeting 6:00 Pm	Project Check In Due	Office hours 3:00 Pm - 5:00 Pm			Finish final concepts and deliverables for the project
Week of 2/17/2025	Team Meeting 1:00 Pm		Office hours 3:00 Pm - 5:00 Pm Team Meeting 6:00 Pm.		Office hours 3:00 Pm - 5:00 Pm			Finish Technical and Real-Problem solutions for the project. Finish 50% of final project draft
Week of 2/24/2025			Office hours 3:00 Pm - 5:00 Pm Team Meeting 6:00 Pm	Team Meeting 6:00 Pm	Office hours 3:00 Pm - 5:00 Pm Team Meeting 7:00 Pm	Final Project Due		Edit and review the final paper. Get feed back and put final touches on paper.

Fig. 17 Schedule of meeting times and deadlines

C. Communication

Communication was a vital part of all aspects of the work done. The team made sure that everyone felt comfortable expressing their own opinion and was heard. The team mostly used a texting group chat for communication. There were also many meetings arranged to ensure that everyone understood the tasks that others were doing. The group also communicated with the teaching assistants during office hours to get confirmation on parts of the project. There was also a group folder shared with everyone so that information and ideas could be easily shared. Within the group folder an AIAA cheat sheet was used to ensure proper AIAA formatting throughout the report.

D. Challenges and Lessons Learned

Throughout the project, the group learned very useful lessons about collaboration and working on longer-term projects. The greatest challenge the group faced was working within each of our different schedules to meet and get tasks done. Each member of the group had tests and other assignments to do while also finding the time to work on this project. This led to the importance of communication and knowing when to prioritize other classes or when to prioritize this project. Having a group calendar and using When2meet was very beneficial in creating meeting times. Another lesson learned was the importance of team planning. In the beginning of the project, the team had very little knowledge of using overleaf or AIAA paper formatting. This led to a learning curve that the team needed to overcome, which took a significant amount of time. Working with each other while coding within the overleaf was very beneficial because we all slowly learned the language together. While learning overleaf was difficult, we eventually created a share knowledge of the software. This led to more efficient work sessions and the work became a little bit easier as we went. We had multiple work sessions together and barely worked on the project by ourselves, this helped pushed collaboration and led to a better final project. In general, we learned that the most important thing when working in a group is communication within the group and prioritizing the project when it is needed.

VII. Conclusion

Upon analyzing the trajectory of the balloon as they been moved from the air, it is clear to see how the atmosphere effects affected it. The graphs show that as the balloon ascended through the air the changes in temperature caused changes in pressure which led to changes in the balloon's volume. As the pressure decreased and less the balloon expanded. This expansion eventually caused the balloon to pop because the balloon latex stretched past its critical stress point. In the real-world problem the problem was proposed as measuring the changes in altitude of a weather balloon as it ascends. The real-world problem also found many different factors that affect the balloon as it ascends. Some examples of these factors are the pendulum motion of the radiosonde or the collection of condensation on the balloon that weighs it down. The academic problem took this problem and simplified it to make it solvable with the skills which were taught in class. With the academic problem, the more complicated realistic factors were ignored because they were too difficult to measure. This led to the problem being labeled a hydrostatic problem. Hydrostatics were discussed thoroughly in class and helped in finding a solution to this problem. Upon solving the problem, it was clear that the changes in pressure were the cause of the buoyant force. This was discussed in class, and more specifically the volume of displaced fluid was the cause of the buoyant force. This was also defined while going through the method part of the technical question. Although both of these problems were successfully solved, it was important for the team to collaborate with each other. This was the task of the Team Manager. The team shared a calendar with office hours, meeting times, and deadlines on it. This was in place to ensure that all members understood what was happening and no one became uninformed. It also was important for constant in person team work sessions so each person on the team could stay up to date with what the others were doing. Throughout our report we found how a weather balloon is affected as it rises through the air and what different factors affect the data that it is gathering. It is also important to look at the costs of these balloons and the different components of them. It is important to continue to study weather balloons because of their importance in weather data collection and learning more about the global warming crisis.

Appendix

A. Use of Artificial Intelligence in this Document

Artificial Intelligence (AI) was used to assist in the formatting of the report in Overleaf. In particular, artificial intelligence was used to use the correct citation format and syntax for web pages, journals, and books. It also proved helpful with tedious tasks such as alphabetizing the terms in the "Nomenclature" section. AI was also helpful in translating equations from their written form to latex syntax.

B. Code

The following pages contain the code used to generate the results, plots, and simulations for this investigations.

```
[ ]: import numpy as np
import matplotlib.pyplot as plt
```

```
[ ]: # reference values
g0 = 9.80665 # m/s
# geopotential altitude
h1 = 11000 # m
h2 = 25000 # m
h3 = 47000 # m
rE = 6.3781e6 # radius of the earth, m
# Ambient Temperature
T0 = 288.16 # K at sea level
T1 = 216.66 # K at 11 km
T2 = T1
T3 = 282.66 # K
# Atmospheric Pressure
p0 = 101325 # Pa at sea level
p1 = 2.2700e4 # at 11km
p2 = 2.4495e3 # at 25km
p3 = 1.1973e2 # at 47km
# Air Density
rho0 = 1.2250 # kg / m^3
rho1 = 0.36480 # kg / m^3 at 11 km
rho2 = 0.039333 # kg / m^3 at 25 km
rho3 = 1.4757e-3 # kg / m^3 at 47 km
R = 287 # J / kg*K
```

```
[ ]: def T(h):
    if h <= 11000: # gradient layer
        return T0 + (T1-T0) / h1 * h
    elif h <= 25e3: # isothermal layer
        return T1
    else: # gradient layer
        return T2 + (T3 - T2) / (h3 - h2) * (h - h2)
def p(h):
    if h <= 11000: # gradient layer
        c = (T1 - T0) / h1
        return p0 * (T(h) / T0)**(-g0/(c*R))
    elif h <= 25e3: # isothermal layer
        return p1 * np.exp(-g0/(R*T(h))*(h-h1))
    c = (T3 - T2) / (h3 - h2) # gradient layer
    return p2 * (T(h) / T2)**(-g0/(c*R))
def rho(h):
    if h <= 11000: # gradient layer
        c = (T1 - T0) / h1
        return rho0 * (T(h) / T0)**(-g0/(c*R)-1)
```

```

elif h <= 25e3: # isothermal layer
    return rho1 * np.exp(-g0/(R*T(h))*(h-h1))
c = (T3 - T2) / (h - h2) # gradient layer
return rho2 * (T(h) / T2)**(-g0/(c*R) - 1)

def hg(h): # returns the geometric altitude at the geopotential altitude
    return h*rE / (rE - h)
def g(hg): # returns the force of gravity at the geopotential altitude
    return g0 * rE / (rE + hg)

```

```

[ ]: # Will use Euler Method -> First Define acceleration
t0 = 0.0002 # m
massPayload = 3.0 # kg
massGas = 0.5 # kilograms
Gas = "He"
if Gas == "H2":
    molarMassGas = 2.016 # H2
    priceGas = 4.825 # dollars per kg
if Gas == "He":
    molarMassGas = 4.0026 # He
    priceGas = 95.120 # dollars per kg
r0 = (3*(massGas*1000/molarMassGas)*8.314*T0/(4*np.pi*p0))**(1/3) # initial
    ↪ radius, m
densityLatex = 920 # kg/m^3
latexVolume = ((r0 + t0)**3 - r0**3)*np.pi * (4/3) # kg / m^3
massBalloon = densityLatex * latexVolume # kg
massTotal = massGas + massBalloon + massPayload

Cd = 0.47
G = 0.000500e9 # Pa Shear Modulus
criticalStress0 = 3.4e6 # Pa critical stress at STP (needs more research)
criticalStress = criticalStress0

```

```

[ ]: def balloonRadius(a, b, c, d, e, f):
    # determines the radius of the balloon from a 5th degree polynomial
    poly = np.polynomial.polynomial.Polynomial([a, b, c, d, e, f])
    roots = poly.roots()
    # Filter to only real roots (using a tolerance for the imaginary part) and
    ↪ non-negative values.
    positive_real_roots = [r.real for r in roots if np.isclose(r.imag, 0,
    ↪ atol=1e-8) and r.real >= 0]
    # Return the smallest positive real root if available; otherwise, return NaN.
    if not positive_real_roots:
        return np.nan
    return min(positive_real_roots)

def proximityToFailure(r, t): # evaluates the proximity to critical stress

```

```

    return criticalStress - G*(r/r0 - r0**2/r**2)

a = np.linspace(0,47000,47001)
diameterValues = np.ones(np.size(a))
temperatureValues = np.zeros(np.size(a))
pressureValues = np.zeros(np.size(a))
densityValues = np.zeros(np.size(a))

for i in range(np.size(a)):
    temperatureValues[i] = T(a[i])
    pressureValues[i] = p(a[i])/1000
    densityValues[i] = rho(a[i])
    diameterValues[i] = 2*balloonRadius(-2*G*t0*r0**4, 0, -3*(massGas/2.016)*8.
↪314e3*T(a[i]), 2*G*t0*r0, 0, 4*np.pi*p(a[i]))
a /= 1000 # convert to km for clear plotting

```

```

[ ]: plt.figure(dpi=1200)
plt.plot(temperatureValues, a, 'k')
plt.xlabel("Temperature (K)")
plt.ylabel("Altitude (km)")
plt.show()

```

```

[ ]: plt.figure(dpi=1200)
plt.plot(diameterValues, a, 'k')
plt.xlabel("Diameter (m)")
plt.ylabel("Altitude (km)")
plt.show()

```

```

[ ]: plt.figure(dpi=1200)
plt.plot(pressureValues, a, 'k')
plt.xlabel("Atmospheric Pressure (kPa)")
plt.ylabel("Altitude (km)")
plt.show()

```

```

[ ]: plt.figure(dpi=1200)
plt.plot(densityValues, a, 'k')
plt.xlabel("Air Density (kg / m^3)")
plt.ylabel("Altitude (km)")
plt.show()

```

```

[ ]: def eulerMethod(t0, massGas, tf, dt):
    time = np.arange(0, tf + dt, dt) # time array
    y = np.zeros(np.size(time)) # altitude array
    vy = np.zeros(np.size(time)) # velocity array
    forces = np.zeros(np.size(time)) # force array
    diameter = np.zeros(np.size(time)) # radius array

```

```

pToF = np.zeros(np.size(time)) # proximity to failure array
# initial conditions:
y[0] = 0 # launch altitude
vy[0] = 0 # launch velocity

success = False

for n in range(len(time) - 1):
    # radius function call for the current altitude
    radius = balloonRadius(-2*G*t0*r0**4, 0, -3*(massGas*1000/2.016)*8.
    314*T(y[n]), 2*G*t0*r0, 0, 4*np.pi*p(y[n]))
    diameter[n] = radius * 2 # log diameter for output

    t = t0 * (r0/radius)**2 # new thickness due to balloon expansion

    criticalStress = criticalStress0 * (1 - .01*(T(y[n])-T0))
    pToF[n] = proximityToFailure(radius, t)
    if pToF[n] < 0:
        print("The Balloon Popped! The altitude is", y[n])
        success = False
        break

    # now calculate forces to find net force on the balloon
    drag = .5*rho(y[n])*Cd*vy[n]*(radius**2*np.pi) # force due to drag on
    the balloon
    Fb = rho(y[n])*(radius + t)**3*np.pi*(4/3)*g(y[n]) # force of buoyancy
    due to pressure gradient
    Fg = - massTotal*g(y[n]) # force of gravity on the balloon
    force = Fb + Fg - drag # net force on the balloon
    forces[n] = force

    ayn = force / massTotal # newton's second law

    vy[n + 1] = vy[n] + ayn * dt # velocity at next timestep

    y[n + 1] = y[n] + vy[n] * dt # position at next timestep

    if (vy[n] < 0 or y[n] < 0): # check if balloon is moving downward
        if y[n] < 350000:
            break
        print("The maximum altitude is ", y[n])
return vy, y, time, forces, diameter, pToF

```

```

[ ]: # USER INPUT:
massGas = 0.2
t0 = 0.0002

```

```

[ ]: # simulation
# initialize data arrays
masses = np.zeros(100)
thicks = t0 * np.ones(100)
heights = np.zeros(100)
maxDiameters = np.zeros(100)
success = [""] * 100
for i in range(np.size(masses)): # limit iterations to 100 to prevent infinite
    loops
        vy, y, time, forces, diameter, proxToFail = eulerMethod(t0, massGas, 20000,
    0.5) # call Euler method
        masses[i] = massGas # store other data
        heights[i] = np.max(y)
        maxDiameters[i] = np.max(diameter)
        if (heights[i] >= 35000): # condition for success
            success[i] = "success"
            break
        else:
            success[i] = "fail"
            massGas += 0.01 # increment the amount of gas used if success is not
    achieved
            massGas = np.round(massGas, 2)
            # recalculate the radius, volume, and mass for this additional gas
            r0 = (3*(massGas/molarMassGas)*8.314e3*T0/(4*np.pi*p0))**(1/3) #
    (3*initialVolume/(4*np.pi))**(1/3) from volume
            latexVolume = ((r0 + t0)**3 - r0**3)*np.pi * (4/3)
            massBalloon = densityLatex * latexVolume # kg
            massTotal = massGas + massBalloon + massPayload

for i in range(2, np.size(time)):
    # remove zero values at the end of the arrays output
    if y[i] == 0:
        vy[i] = np.nan
        y[i] = np.nan
        forces[i] = np.nan
        diameter[i] = np.nan
        time[i] = np.nan
        proxToFail[i] = np.nan
    # trim other data arrays
    masses = np.trim_zeros(masses, 'b')
    cost = np.round(masses * priceGas, 2)
    success = success[:np.size(masses)]
    thicks = np.resize(thicks, np.shape(masses))
    maxDiameters = np.round(np.trim_zeros(maxDiameters, 'b'), 1)
    heights /= 1000 # covert to km
    thicks *= 1000 # convert to mm
    heights = np.round(np.trim_zeros(heights, 'b'), 2)

```

```
[ ]: # Plotting
y /= 1000 # convert to km
time /= 60 # convert to minutes
plt.figure(dpi=1200)
plt.plot(time, y, 'k')
plt.xlabel("Time Elapsed (min)")
plt.ylabel("Altitude (km)")
plt.show()
```

```
[ ]: plt.figure(dpi=1200)
plt.plot(time, vy, 'k')
plt.xlabel("Time Elapsed (min)")
plt.ylabel("Velocity (m/s)")
plt.show()
```

```
[ ]: plt.figure(dpi=1200)
plt.plot(time, forces, 'k')
plt.xlabel("Time Elapsed (min)")
plt.ylabel("Net Force (N)")
plt.show()
```

```
[ ]: plt.figure(dpi=1000)
plt.plot(time, diameter, 'k')
plt.xlabel("Time Elapsed (min)")
plt.ylabel("Balloon Diameter (m)")
plt.show()
```

```
[ ]: proxToFail /= 1000000
plt.figure(dpi=1000)
plt.plot(y, proxToFail, 'k')
plt.xlabel("Altitude (km)")
plt.ylabel("Proximity to Failure (MPa)")
plt.show()
```

```
[ ]: import pandas as pd
```

```
[ ]: # create a table of outputs
df = pd.DataFrame(np.transpose(np.array([masses, cost, thicks, maxDiameters,
→ heights, success])), index=None, columns=["m He (kg)", "Cost ($USD)", "t0_
→ (mm)", "D (m)", "a (km)", "success?"])
```

```
[ ]: df.to_csv('simulation.csv', index=None)
```

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